KAVIN BALAMURUGAN

Software Engineer | Full-Stack & AI Applications

+91 6383587148 • kavin.balamurugan2k@gmail.com • Chennai (Open to Relocation) • linkedin.com/in/vakbkavin

PROFESSIONAL SUMMARY

Software Engineer with **5+ years** of experience at Cognizant building enterprise applications, frontend platforms, backend services, and AI-powered solutions. Hands-on experience across Angular, TypeScript, Python, ASP.NET Core, microfrontend architecture, Backend-for-Frontend systems, and enterprise Generative AI integration. Experienced in developing end-to-end application features, integrating specialized AI agents, engineering secure API communication, and contributing to cloud-based application architectures. Combines strong frontend engineering expertise with practical backend and AI application development experience.

TECHNICAL SKILLS

Frontend	Angular 13–22, TypeScript, JavaScript ES6+, React, Next.js, RxJS, HTML5, SCSS
Architecture	Microfrontends, Module Federation, Native Federation, BFF Pattern, REST APIs, Distributed Application Integration
Backend	Python, FastAPI, ASP.NET Core, C#, Node.js
AI & GenAI	LLM Integration, Azure AI Foundry, Gemini, Vertex AI, RAG, Natural Language Query, AI Agents, A2A, A2UI
Cloud & Data	Azure, GCP, AKS, GKE, Snowflake, Azure Table Storage, Azure Blob Storage, PostgreSQL
Security	OIDC, OAuth Concepts, SSO, AES-GCM, Antiforgery, API Governance, Secure API Integration
Tools	Git, Docker, Kubernetes, Azure DevOps, Vite, Webpack, Storybook, Postman, GitHub Copilot

PROFESSIONAL EXPERIENCE

Associate – Frontend & AI Engineer *January 2024 – Present*
Cognizant Technology Solutions

1. Marketing Assistant – Enterprise Marketing AI Platform

Technologies: Angular, TypeScript, JavaScript, Vite, ASP.NET Core, Python, Azure AI Foundry, Snowflake, Microfrontends, A2A

Campaign Assistant evolved into Marketing Assistant, a conversational platform integrating marketing workflows and specialized agents, including BRD Generator and Audience Builder, with future capabilities such as MBO and Narrative Agent. NeuroSan coordinates agent invocation and user communication.

- Developed **Audience Builder (AB)** end-to-end, migrating the application from Python Dash to a JavaScript and Vite-based microfrontend, and integrating it into the enterprise Marketing Assistant platform.
- Engineered an **ASP.NET Core Backend-for-Frontend (BFF)**, implementing controllers, services, and filters to coordinate secure communication between frontend applications and backend services.
- Reduced Audience Builder's initial load time by **40%** through its migration and frontend integration.
- Implemented secure cross-cluster API communication using reverse-proxy architecture, APIGEE onboarding, **AES-GCM** encryption, and ASP.NET Core Antiforgery validation.
- Built a Natural Language Query pipeline using Azure AI Foundry and a Dynamic Semantic Registry backed by Azure Table Storage, enabling business users to query Snowflake datasets using natural-language terminology.
- Contributed significantly to the broader Marketing Assistant ecosystem, integrating specialized marketing capabilities and collaborating across frontend, backend, AI orchestration, platform, and security teams.
- Contributed to the platform's evolution towards a unified conversational marketing experience, supporting capabilities involving specialized agents, generated artifacts, structured tables, Workfront task creation, and no-code and pro-code agent integrations.

2. Enterprise Generative AI Content Suite

Technologies: Angular, ASP.NET Core, Azure AI

- Developed production AI-powered modules including multilingual translation, persona generation, SEO optimization, AI writing assistance, and pitch creation using **Angular** and ASP.NET Core.
- Contributed across the feature lifecycle, including frontend development, backend integration, testing, and deployment support.
- Delivered stakeholder demonstrations and supported adoption across multiple marketing business units, receiving internal recognition through Cognizant Cheer Points.

3. Cognizant.ai – Enterprise AI Application

Technologies: Next.js, React, TypeScript, Python, FastAPI, Gemini, Vertex AI Search, PostgreSQL, GCP, Cloud Run

- Developed and enhanced frontend and backend features for an enterprise AI-powered application using Next.js, React, TypeScript, Python, and FastAPI.

- Worked across the RAG application lifecycle, integrating Gemini and Vertex AI Search capabilities with the application experience.
- Implemented and validated application features involving frontend-backend integration, cloud storage, and application workflows.
- Designed and evaluated persistent conversation and session architecture using PostgreSQL, including database models and application integration requirements.
- Conducted technical audits and engineering improvements covering authentication, authorization, API security, configuration management, and deployment readiness.
- Identified architectural and production-readiness gaps, including persistence, access control, and deployment concerns, and developed recommendations for subsequent implementation.

4. Media Budget Optimizer (MBO) – Early-stage Initiative

Technologies: Python, Business Rules, Data Requirements, Marketing Analytics

- Collaborating with business stakeholders to establish historical campaign data and benchmark requirements for an enterprise media budget optimization initiative.
- Analyzing channel taxonomies, budget allocation rules, performance benchmarks, and business requirements to inform deterministic optimization engine design.
- Identifying unresolved business decisions and external data dependencies, including Oracle historical data and benchmark availability, before production engine development.

EARLIER EXPERIENCE

Programmer Analyst (Trainee → Analyst) *October 2021 – December 2023*

Cognizant Technology Solutions

Technologies: JavaScript, Angular 14, SCSS, Adobe Experience Manager, Handlebars

- Delivered dynamic frontend components for **Cognizant.com** and developed Angular 14 proof-of-concept applications using the Topaz Design System for enterprise clients including Pearson and Warner Bros.
- Contributed to migrating a legacy healthcare portal to an Experience Design System (EDS)-based architecture, improving page load times by 35% and supporting WCAG 2.1 AA accessibility compliance.

Creative Asset Management Platform (CAMP)

Technologies: JavaScript, Adobe Experience Manager, Webpack, Python, Azure Blob Storage

- Developed reusable frontend components using Adobe Experience Manager and Webpack, reducing developer onboarding effort by over 50% across the component library.
- Integrated Azure Blob Storage with Python APIs to support scalable digital asset workflows.
- Authored component standards and documentation to improve consistency and reuse across development teams.

SELECTED TECHNICAL PROJECTS

A2UI / A2A – Dynamic AI Interface POC (Proof of Concept)

Technologies: Lit, Vite, TypeScript, FastAPI, A2UI

- Developed a proof of concept for agent-driven dynamic user interfaces using Lit, Vite, TypeScript, and FastAPI.
- Integrated A2UI packages and implemented a mock agent transport to generate and deliver UI messages to the frontend.
- Explored agent-to-agent communication patterns and the integration of AI agent responses with dynamically rendered user interfaces.
- Validated the implementation through frontend builds and backend automated tests.

EDUCATION

Bachelor of Engineering – Mechanical Engineering *2017–2021*

Kongu Engineering College | GPA: 7.8/10 | **Languages:** English, Tamil

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- AWS Certified Cloud Practitioner
- GitHub Copilot
- Build with Gemini – Completed Workshops
- Claude Certified Architect – Currently Pursuing